

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE CODE: CSA51 **COURSE NAME:** Cryptography and Network Security

COURSE OUTCOME COVERED

CO3: Examine cryptographic hash algorithms, digital signature schemes, and assess Kerberos and X.509 authentication frameworks for ensuring integrity, authentication, and non-repudiation.CO (BL4 , BL5)

Assessment Tool Weightage: 10%

QUESTIONS: 5

TOTAL MARKS: 50

**Annexure A
MCQ**

Code of Conduct:

I M.Sabeer Khan / 192212101 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M. Sabeer Khan

Annexure A
MCQ

MCQ (Additional Set)

1. Which property ensures the same input always produces the same hash?
A) Avalanche effect
B) Determinism
C) Collision resistance
D) Diffusion
2. What is the primary purpose of SHA-3?
A) Faster encryption
B) Alternative secure hashing standard
C) Data compression
D) Key exchange
3. Which attack tries all possible inputs to match a hash?
A) Replay attack
B) Brute-force attack
C) MITM attack
D) DoS attack
4. What is a salted hash?
A) Encrypted hash
B) Hash with random data added
C) Shortened hash
D) Reversed hash
5. Which algorithm is widely used for password hashing?
A) AES
B) RSA
C) bcrypt
D) DES
6. What does second pre-image resistance ensure?
A) Same input produces same output
B) Hard to find different input with same hash
C) Fast hashing
D) Small hash size
7. What is the output size of SHA-512?
A) 256 bits
B) 128 bits
C) 512 bits
D) 1024 bits
8. Which technique slows down brute-force attacks?
A) Encryption
B) Key exchange
C) Key stretching
D) Compression
9. What is the purpose of a nonce?
A) Encryption
B) Prevent replay attacks
C) Compress data
D) Generate keys
10. Which function is irreversible?
A) Encryption
B) Hashing

- C) Compression
- D) Encoding
- 11. Which algorithm is used in DSS?
 - A) AES
 - B) DSA
 - C) DES
 - D) RC4
- 12. What is required to verify a digital signature?
 - A) Private key
 - B) Public key
 - C) Session key
 - D) Shared key
- 13. What does a digital signature guarantee?
 - A) Confidentiality
 - B) Integrity and authenticity
 - C) Compression
 - D) Speed
- 14. What happens if message digest changes?
 - A) Signature remains valid
 - B) Verification fails
 - C) Encryption fails
 - D) Speed increases
- 15. Which attack involves forging a signature?
 - A) Replay
 - B) Forgery
 - C) DoS
 - D) Sniffing
- 16. What is the role of a hash in signatures?
 - A) Encrypt data
 - B) Reduce size
 - C) Ensure integrity
 - D) Store keys
- 17. Which key is kept secret in signatures?
 - A) Public key
 - B) Private key
 - C) Session key
 - D) Shared key
- 18. What ensures trust in certificates?
 - A) Encryption
 - B) Certificate Authority
 - C) Compression
 - D) Hash size
- 19. What is certificate revocation?
 - A) Key generation
 - B) Canceling validity
 - C) Encryption
 - D) Compression
- 20. What is OCSP used for?
 - A) Encryption
 - B) Certificate validation
 - C) Compression
 - D) Routing
- 21. What is the main goal of Kerberos tickets?
 - A) Encryption
 - B) Authentication

- C) Compression
- D) Routing
- 22. Which protocol uses tickets for authentication?
 - A) SSL
 - B) Kerberos
 - C) FTP
 - D) HTTP
- 23. What is a session key?
 - A) Permanent key
 - B) Temporary key
 - C) Public key
 - D) Private key
- 24. What is mutual authentication?
 - A) One-way authentication
 - B) Both parties verify each other
 - C) Anonymous login
 - D) Weak authentication
- 25. What ensures freshness in Kerberos?
 - A) Encryption
 - B) Timestamp
 - C) Compression
 - D) Hashing
- 26. What is a replay attack?
 - A) Data modification
 - B) Resending valid data
 - C) Encryption
 - D) Compression
- 27. What is the function of an authenticator in Kerberos?
 - A) Data storage
 - B) Identity proof
 - C) Encryption
 - D) Compression
- 28. What happens if system clocks differ in Kerberos?
 - A) Faster authentication
 - B) Failure
 - C) Better security
 - D) No effect
- 29. What is the KDC composed of?
 - A) Client & Server
 - B) AS & TGS
 - C) Firewall & IDS
 - D) Router & Switch
- 30. What is the AS in Kerberos?
 - A) Application Server
 - B) Authentication Server
 - C) Access Server
 - D) Admin Server
- 31. What is X.509 certificate format used in?
 - A) Networking
 - B) Digital certificates
 - C) Compression
 - D) Routing
- 32. What identifies the certificate owner?
 - A) Public key
 - B) Subject field

- C) Signature
- D) Hash
- 33. What ensures certificate integrity?
 - A) Encryption
 - B) Digital signature
 - C) Compression
 - D) Routing
- 34. What is a root certificate?
 - A) Expired certificate
 - B) Self-signed trusted certificate
 - C) Invalid certificate
 - D) Temporary certificate
- 35. What is certificate pinning?
 - A) Storing keys
 - B) Binding certificate to host
 - C) Encryption
 - D) Compression
- 36. What is the risk of weak CA validation?
 - A) Faster speed
 - B) MITM attacks
 - C) Compression
 - D) Routing
- 37. What is the purpose of trust chain?
 - A) Encryption
 - B) Validate certificate hierarchy
 - C) Compress data
 - D) Routing
- 38. What is digital certificate used for?
 - A) Compression
 - B) Authentication
 - C) Routing
 - D) Storage
- 39. What ensures non-repudiation?
 - A) Encryption
 - B) Digital signature
 - C) Compression
 - D) Routing
- 40. What is a hash collision?
 - A) Same input
 - B) Same output for different inputs
 - C) Encryption
 - D) Compression
- 41. Which hash is strongest?
 - A) MD5
 - B) SHA-1
 - C) SHA-256
 - D) CRC
- 42. What is a one-way function?
 - A) Reversible
 - B) Irreversible
 - C) Fast
 - D) Small
- 43. What ensures authentication in PKI?
 - A) Encryption
 - B) Certificates

- C) Compression
- D) Routing
- 44. What is key escrow?
 - A) ~~Key deletion~~
 - B) Key storage by third party
 - C) Encryption
 - D) Compression
- 45. What is identity spoofing?
 - A) ~~Encryption~~
 - B) Fake identity
 - C) Compression
 - D) Routing
- 46. What is trust anchor?
 - A) Root CA
 - B) Client
 - C) Server
 - D) Router
- 47. What ensures integrity of message?
 - A) ~~Encryption~~
 - B) Hash
 - C) Compression
 - D) Routing
- 48. What is digital envelope?
 - A) Hash
 - B) ~~Encrypted message + key~~
 - C) Compression
 - D) Routing
- 49. What is certificate fingerprint?
 - A) ~~Encryption~~
 - B) Hash of certificate
 - C) Compression
 - D) Routing
- 50. What is PKI used for?
 - A) ~~Routing~~
 - B) Secure communication
 - C) Compression
 - D) Storage